

PowerLED Indoor Lights

- Extremely Low power consumption
- Long life (in excess of 50,000 hours)
- Models with light output to match conventional indoor lights
- No maintenance required: very low Lifetime Ownership Cost



BINAY - Lighting Up The Future



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INVENTING NEW TECHNIQUES OF PRODUCING LIGHT

BINAY LED-based Lights

**PowerLED
Street
Light**



**PowerLED Hight Bay Light
PowerLED Strip Light**



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MCUs require twice the memory of 8-bit MCUs, but a 16-bit architecture can actually require less code than an 8-bit architecture. Besides, 16-bit MCUs usually execute tasks more quickly. To manage 10-bit ADC data or applications that require 16-bit math, 8-bit MCUs require CPU overhead. Further, many MCU offerings today have a single working file, or accumulator, through which the data must be moved to be acted upon, requiring excess CPU overhead compared to register-based architectures.

Transition and recovery: low-power mode

Running the MCU at full speed all the time will never lead to a truly low-power design, even if the lowest-power MCU is used. The biggest power savings are possible only by frequently switching the MCU to a low-power sleep state from normal state and vice-versa.

Transition to low-power modes. The commands available for CPU transition into the LPMs are a very important factor when choosing any MCU. The MCU is switched into LPMs by configuring bits in status and control registers, i.e., by issuing a sleep command over the bus.

Recovery or wakeup from low-power modes. The wakeup request is similar to interrupt requests, except that the CPU operation is stopped. Unlike interrupt requests, wakeup requests are accepted regardless of processor status in the PS register. Hence wakeup is possible even if interrupts are disabled or interrupt priority is not defined.

To recover or wake up from LPMs, an event from a resource is required. The possible resources that can wake up a controller depend on the type of LPM, as shown in Fig 5. A good low-power MCU has extensive interrupt capability, providing interrupts for all its peripherals and many external interrupts for external events.

Recovering from LPMs into normal operating modes can impose a significant delay. Therefore a key feature to look for in any MCU is the shortest recovery time. MCU recovery time from some LPMs should be fast enough to meet the response times of interrupts. Total T_{recovery} is approximately 2.5 ms with T_{wakeup} of 1.4 ms, PLL stabilisation time of 1 ms and gear-up time of about 75 μ s. This illustrates the ability to optimise the design for low power while having a flexible architecture in the MCU that helps to achieve high performance and lower power consumption.

To be concluded next month

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